AI-Based Skin Allergy Detection and Personalized Healthcare Recommendation System Using Deep Learning

Poojasree P D
Dept of computer science and engineering
Vels institute *of science,technology and advanced studies*
chennai,India
poojadesulu10@gmail.com

Kaviyashree K
Dept of computer science and engineering
Vels institute *of science,technology and advanced studies*
chennai,India
kaviyashree605@gmail.com

Janani R
Dept of computer science and engineering
Vels institute*n of science,technology and advanced studies*
chennai,India
jananiramesh728@gmail.com

S.Divya Bairavi
Dept of computer science and engineering
Vels institute of science,technology and advanced studies
chennai,India
divyabairavi.se@vistas.ac.in

Abstract—Many skin conditions such as urticaria, eczema, psoriasis and contact dermatitis look similar in photos and are challenging to differentiate for a non-specialist. This paper presents a prototype, called DermaAI that brings together an EfficientNetV2-B0 image classifier with a complete patient-facing workflow, as opposed to a single predicted label. After a user uploads an image of their skin condition, together with some basic details and symptoms, they receive the most probable diagnosis with high level of confidence, with an overlay Grad-CAM visualization, a rule-based severity assessment via 6 symptom sliders, relevant food and clothing recommendations, a downloadable PDF-report and information about the location of dermatologists nearby. The interface is in both English and Tamil and it allows the symptoms to be read out loud. The current form of the classifier can take 4 classes of input that are 224x224 in a range of RGB values with an accuracy of 85.80% on the test set which must then be evaluated on a test set held out for the conference. The purpose of the system is to support the informal screening and patient education and does not take the place of a dermatologist's care. Abstract— Skin allergy detection algorithms incorporate several approaches, such as deep learning, transfer learning, image classification, Gradient-weighted Class Activation Mapping (Grad-CAM), explainable AI, personalized healthcare recommandation, severity assessment and Streamlit.Skin allergy detection algorithms employ a variety of techniques, including deep learning, transfer learning, image classification, deep-learning-inspired model explanations like Gradient-weighted Class Activation Mapping (Grad-CAM), individualization of healthcare recommendations, severity assessment, and Streamlit.

I.INTRODUCTION

Skin and subcutaneous conditions are associated with a negative effect on quality of life even in non-life-threatening cases. Urticaria, eczema, psoriasis, contact dermatitis – these four diagnoses have similar appearances in photos, and a patient is unlikely to distinguish between them based on visuals alone. Meanwhile, an appointment with a dermatologist may be scheduled weeks in advance, and in the meantime, the patient may research their symptoms online or seek informal advice from friends and family. An application that formalizes and structures this initial interaction, providing a specific diagnosis and next steps, is valuable to both the patient and their doctor.

While image classification is a routine task in medical imaging, especially after the advent of large convolutional networks [4], a lone predicted label is insufficient to most users. A patient researching their skin issue is unlikely to be satisfied with a mere label; they are more likely to benefit from an assessment of prediction confidence, an explanation of what the model saw in the image, an evaluation of their symptoms, an indication of severity, and recommendations for further steps. DermaAI is a prototype that offers all of these features: an EfficientNetV2-B0 classifier [5] is complemented with Grad-CAM visualization, a rule-based severity assessment, personalized recommendations, multilingual support, an automatically generated report, and a directory of local dermatologists. The application is built as a single connected unit rather than a standalone classifier to better serve the needs of the user at the first point of contact with a suspected skin condition. The motivation for this approach comes from the observation that very few patients will contact a dermatologist immediately upon noticing a skin anomaly. Instead, they are more likely to look at the image, possibly consult a friend or family member, and only then – schedule an appointment with a specialist. A system that recognizes this and addresses its capabilities to the initial, informal interaction has a much better chance of impacting the patient's quality of care.

In short, this work provides:(1)a four-class skin condition image classifier based on EfficientNetV2-B0;(2)Grad-CAM explanation of the image classification;(3)a rule-based symptom severity assessment that operates alongside the image classifier;(4)personalized diet, clothing and supportive-care recommendations based on the classification;(5)multilingual (English/Tamil) interface with speech-to-text support;(6)an

automatically generated patient report with classification, confidence assessment, and image attached.

II. PROBLEM STATEMENT

Urticaria, eczema, psoriasis and contact dermatitis are four skin conditions that are challenging to differentiate from one another, especially for a non-specialist. Dermatology consultations are not always readily available, and patients are likely to turn to informal resources in the meantime, such as searching their symptoms on the web. Most existing medical image classifiers offer little beyond predicted labels; there is a need for a system that formalizes and structures a patient's initial interaction with a suspected skin condition.

Specifically, the following features are required– image classification with explanation of the model's inner workings (e.g. Grad-CAM)– symptom assessment with defined levels of severity– personalized recommendations for diet, clothing, supportive care– multilingual (English/Tamil) interface with voice input;– automatic report generation with personal details, predicted diagnosis, confidence level and image attachment– directory of nearby dermatologists for consultation scheduling assistance.

III. OBJECTIVES

- Build an EfficientNetV2-B0-based classifier that can differentiate between Contact Dermatitis, Eczema, Psoriasis and Urticaria from a single uploaded image.
- Add a visual explanation of the classification in the form of a Grad-CAM overlay.
- Implement a rule-based severity assessment with 6 symptom indicators that operate alongside the image classifier.
- Personalize food, clothing and supportive-care recommendations based on the predicted diagnosis, rather than using a generic list.
- Make the application accessible to a wider audience by implementing a bilingual interface with voice input.
- Automate the generation of a PDF patient report that includes personal details, the predicted diagnosis, confidence assessment and the uploaded image.
- Provide consultation assistance by locating nearby dermatologists and finding those that accept the user's insurance.

IV. RELATED WORK

Convolutional networks are the established solution in image recognition tasks, and several works have demonstrated the efficacy of ensembling or single CNNs in classifying skin lesions [3]. EfficientNet's scaling strategy has seen broad adoption due to its accuracy-per-parameter gains over manually designed CNNs [5]; while larger variants of the network see use in research, the lightweight B0 version is a good fit for this application as a student project outside of a research cluster. Nawale and Gade's implementation of an EfficientNet-based classifier with an integrated clinical decision support system is the closest to this work, and served as inspiration for its design [1].

A competing line of research concerns interpretability of vision models, especially in a clinical setting. A softmax prediction is rarely informative on its own, and Han et al. highlight deep learning's potential to inform – rather than replace – physicians by offering transparent explanations [2]. DermaAI implements such transparency in the form of a Grad-CAM overlay for every prediction, allowing the user to evaluate what parts of the image contributed to the diagnosis. This serves as a diagnostic aid to the physician and a trust-building measure for the patient – the model is not presented as an oracle, but as a classifier with specific features it pays attention to. Similar approaches to medical image classification are limited, with most consumer-facing applications focusing on the predicted label and optionally its confidence score.

A significant number of consumer-oriented applications address skin diagnosis as an isolated task, whereas DermaAI treats it as one step in a larger process. The predicted diagnosis is immediately put to use in determining the symptom severity and suggesting follow-up steps, which is more efficient than post hoc analysis. Symptom assessment follows a rule-based, rather than machine-learned approach for simplicity and to allow clinical oversight – the rules can be reviewed by a physician, whereas a complex severity assessment model would require retraining. DermaAI's primary innovations over related works [1], therefore, concern the practical application of the predicted diagnosis. Symptom-driven severity assessment, voice input, and location-based dermatologist search are features that were not present in earlier applications and add to the utility of the classifier.

Table III summarizes the features provided by a typical skin-checking app, and compares them to the features available in DermaAI . With the exception of image classification itself, none of the listed capabilities are common to both.

TABLE III
FEATURE COMPARISON WITH TYPICAL SKIN-CHECKER APPS

Feature	**Typical skin-checker app**	**DermaAI (proposed)**
Image classification	Yes	Yes (using EfficientNetV2-B0)
Visual explanation of prediction	Rare	Grad-CAM heatmap
Symptom-based severity score	No	Rule-based, 6 indicators
Personalized food/clothing advice	No	Condition-specific
Bilingual + voice input	Rare	English & Tamil
Auto-generated patient report	Rare	PDF with image & confidence
Nearby-dermatologist lookup	No	Location-based search

V. PROPOSED SYSTEM

Fig. 1 shows the architecture of the proposed system, which is designed as a connected set of modular units. The application begins with patient information and symptom description, followed by the image upload and processing steps. The processed image is then analyzed alongside the symptom information by the AI prediction module and severity assessor, with the assessment results feeding into recommendations, report generation and the dermatologist directory.

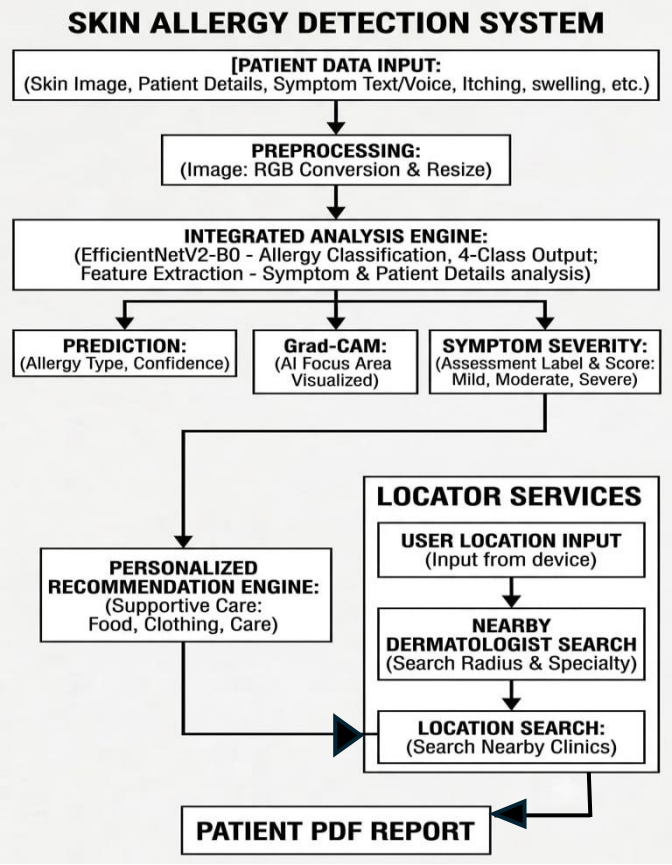


Fig. 1. System Architecture of the proposed skin allergy detection and personalized healthcare recommendation system.

VI. SYSTEM MODULES

The application is separated into 10 modules for easier development, as shown in Table II. Each of the modules serves a particular function, and the division allows for greater flexibility when updating the user-facing features (patient report, language/voice interface) or adjusting the rule-based severity assessment or the learned model.

TABLE II

APPLICATION MODULES

Module	What it does
Patient Details	Collects name, age, gender, contact details, previous allergy history and current symptoms prior to analysis.
Skin Image Upload	Takes a JPG/JPEG/PNG image, converts it to RGB and resizes it to 224×224.
AI Prediction	Analyzes the image using EfficientNetV2-B0 and predicts the skin condition with confidence.
Grad-CAM Explainability	Highlights the image regions used by the model for prediction using Grad-CAM.
Severity Assessment	Analyzes the six symptom indicators and assigns the symptom severity score.
Voice Symptom Input	Provides an option to speak symptoms in English or Tamil rather than type them.
Recommendation	Gives diet, clothing and supportive care tips based on the predicted condition.
Patient Report	Generates a PDF report with patient details, predicted diagnosis, confidence score and image.
Nearby Dermatologist	Finds dermatologists or skin clinics in the region with the option to sort by distance.
Language / UI	Switches between English and Tamil interface, manages user input/output.

VII. METHODOLOGY

(A) Image preparation and categorization: The images are copies and categorized (A). This file is read into a PytPrty image, and converted to The RGB colorspace image was resized to 224×224 and converted. Being able to export to a number array for modelling input. The output layer of the model is made up of 4 nodes, representing four classifications; the predicted label is found using the argmax for the output. The predicted confidence score is given as a string, and the vector is returned. The "vector" is returned and confidence score is given as a string.

B. EfficientNetV2-B0: The compound scaling approach of Deep Scale.B. EfficientNetV2-B0: The Deep Scale 'compound scaling approach' of B. As for the design of the efficientNet, this is used to speed up the training of the model [5]. This lower number of parameters, when compared to the standard CNN, brings great advantages. architectures. The B0 variant becomes the variant chosen for this implementation. Because of its positive ratio of size to performance – Larger variations of the network do exist, but work better in this context. Distributed training of a Cluster. The classifier is trained using Use of a pre-trained vision model to adapt to, known as transfer learning. was not trained from the 4-class discrimination task, but scratch. When model training, the learned weights are saved to This is in a format that is specific to Keras and it should be loaded as a reusable resource at App start-up using a caching tool.

C. Explainable Prediction: After prediction, the selected label is passed to Grad-CAM, which generates a relevancy heatmap based on the model's intermediate features. The heatmap is overlaid onto the original image for visualization, allowing the user to gauge what image features contributed to the diagnosis. This approach to post hoc explanation is appropriate for this setting, as it highlights the evidence for the prediction without attempting to replace the doSmain knowledge of a dermatologist [2].

D. Severity Assessment: Along with the image, the app collects 6 symptom indicators using sliders, as shown in Fig. ?. The indicators relate to itching, redness or irritation, swelling, the extent of the affected area, pain or discomfort and open skin, blistering or oozing. These are evaluated by a rule-based severity assessment module alongside the predicted diagnosis to advise the next course of action. By operating independently of the image classification, the severity assessment has lower barriers to clinical validation and provides a degree of separation from potential biases in the learned model.

E. Personalized Support: The predicted diagnosis is used by the recommendation engine to suggest relevant diet, clothing and supportive-care options, which are provided in a tabbed interface for ease of navigation. The same information is compiled into a PDF patient report along with the personal details and the recommended next steps, serving as a useful takeaway for the user. Meanwhile, the location-aware dermatologist directory finds medical professionals and clinics in the region, assisting the patient in scheduling a follow-up consultation.

F. Dataset Considerations: A skin condition image classifier Usually needs a labeled picture database that contains the names of the pictures. The photo(s) for this application should comprise the 4 target inland, with part of them kept for future use, Requires validation and testing to get an unbiased estimate of accuracy. Care needs to be taken to indicate the source of each The informed consent is required in conjunction with the terms of the use of the picture. Wappy status any patient pictures. If real people are publicly portrayed in photographs and you identify them in that medium, Categorically, if they are used, they need to be processed according to the relevant, privacy laws. After preprocessing the data-set, the model training can be performed in a Due to the high cost of CPUs, distributed setting on graphics processing units to reduce the The time taken to perform this task. Sliac has integrated the core application, however, but it does not require any additional software. needs a lot of CPU power to operate, a midrange CPU is There is enough data for training and prediction purposes. Contribute the software aspects, You need a recent version of Python, for the application. TensorFlow/Keras, Streamlit, Pillow, SpeechRecognition and All these can currently be found on PyPI, ReportLab. An active you'll need a connection to the internet, too.

VIII. SYSTEM REQUIREMENTS

The compiled prototype needs to be run using Python 3.9+. To perform the classification, I've decided to implement the model using TensorFlow/Keras, and present this model interface on a new user interface—Streamlit. interface, Pillow to process the input image, SpeechRecognition ReportLab to create the to create evergreen html, for voice to text. patient report (requirements.txt). For the dermatologist location To use search, you will need some sort of internet access. On the hardware On the other hand this prototype allows running on a “mid-range” CPU, Acceleration of GPUs should be recommended for faster though. performance and training. User should have a cell phone or a computer and a camera. and internet access. Moreover, the conversion of speech into text is also available. Needs to have a plug-in microphone that is easily accessible.

IX. IMPLEMENTATION

The prototype was implemented in Python and Streamlit is used to build the prototype. the main framework, TensorFlow/Keras for deep learning, and canary for bug testing.the main framework, deep learning TensorFlow/Keras, and canary for bug testing. model prediction, Pillow for image processing, For voice input: SpeechRecognition, for pdf output: ReportLab A YouTube player for iOS as well as an entire suite of libraries that support HTTP and geolocation. requests. Patient information and prognosis is stored in Streamlit’s session state throughout the request-response cycle. The interface is bilingual and can be alternated between languages: modifying the appropriate labels and possible input options. The To recognize Tamil, voice to text component first tries to recognize Tamil. If a person says nothing Tamil, reverts to Indian English. phonemes are detected.

In the user's interface it is broken down into panels, shown in Fig. 1. The user starts their interaction with the app by selecting their preferred language, entering their personal details, along with a description of what their current symptoms are. They move on to estimating the extent of those symptoms as represented by the 6 labelled on the Symptom Severity panel, use the sliders to adjust how severe the symptoms are. As soon as they are ready to proceed, takes a picture of their skin, and it lists the skin condition on the Skin page. A picture of their skin condition is uploaded to the Skin page when they click proceed.

The Image Upload panel will launch the AI Analysis and Grad-CAM Explainability panels. The AI Analysis panel will be displayed. The confidence score in percentages when predicted diagnosis is "yes", a short statement on the certainty of the confidence level (high/moderate/low). Grad-CAM Explainability adds the original image to using the generated heatmap to give some background information to the diagnosis. The CHI's score is not used to represent confidence, but rather it is used as a visual descriptor of it; and should NOT be viewed as an absolute value.

The typical users' interaction then follows a linear path through the i) panels as above. Following uploading an image, firstly, analysis starts within a few seconds, and typically finishes in another few seconds. The user is then able to view the diagnosis, confidence score, and the image using Grad-CAM overlaid. From there, they can then oversee/review: Diet, Clothing and Supportive Care recommendations in tabs, local dermatologist's recommendations, click Save, save the Patient Report as a PDF and take it with them to the Everyday Event appointment.

On any occasion, the user should not be expected to have content knowledge of the operation of convolutional networks; simplicity is achieved by abstracting off the complexity that puts the person on a process and provides them with actionable next steps.

X. USAGE OF REFERENCES IN THIS WORK

This section serves to detail the usage of each of the five references included in the work, specifically, which part they appear in and what role they play in support of the paper's argumentation.

Nawale and Gade's work [1] was the inspiration for this one, hence it appears in the Related Work section in the comparison with the existing solutions. The closest counterpart to DermaAI is their EfficientNet-based classifier with clinical decision-support layer, which inspired the general pipeline of our own solution. We cite it again while elaborating on the differences, namely, the symptom-severity engine, the bilingual voice assistant and the dermatologist lookup feature. Han et al.'s paper [2] is leveraged to justify the need for explainability; we quote it while discussing the softmax scores in the Related Work section, and use the Grad-CAM technique again in the Methodology, Explainable Prediction subsection to show how the heatmap can complement the confidence score in the interface. Esteva et al. [4] appears in the Introduction to provide background on the state-of-the-art in image classification for dermatology, specifically, to highlight the achievement of the dermatologist-level accuracy for skin-lesion photography published the same year our work was conceived – to show how limited such a score is for the patient triage scenario. Harangi's survey [3] is used in the Related Work section to discuss the trends in CNN-based skin-lesion classification, particularly, the accuracy improvements afforded by ensembling and increased model capacity, which is why a single EfficientNet is chosen for this project over more complex alternatives. Finally, Tan and Le's paper [5] is used twice, once in the Related Work, once in the Methodology, subsection EfficientNetV2-B0, to discuss the compound scaling of the EfficientNet architecture and justify the choice of the B0 width multiplier specifically.

XI. RESULTS AND DISCUSSION

The results are presented in Table I below for the current configuration of the prototype, which demonstrates a recorded test accuracy of 89.80%. This number will have to be re-evaluated on a held-out test subset before final submission; accuracy is a misleading metric for a medical classifier, and a detailed class-wise analysis with precision, recall, F1, confusion matrix, sensitivity, specificity, and confidence calibration is required before any meaningful conclusions can be drawn about the model's performance.

TABLE I

CURRENT PROTOTYPE CONFIGURATION

Parameter	**Value**
Backbone	EfficientNetV2-B0
Input size	224×224 RGB
Output classes	4 (CD, Eczema, Psoriasis, Urticaria)
Framework	TensorFlow/Keras
Interface	Streamlit web application
Languages	English, Tamil
Reported test accuracy	89.80% (to be re-verified)

Fig. 2 provides a context for the predicted confidence of the model: it shows the output of the running application on four test uploads, which have been processed for the disease label and confidence score. On the image of the urticaria lesion (raised, linear weal on the forearm), the model produced an 80.00% confidence score of Urticaria, which the interface deems moderate; the other three tests have scored over 90.00% and are therefore read as high-confidence predictions. A scaly well-defined plaque was correctly identified as Psoriasis at 95.02% confidence, an inflamed papular patch of skin as Eczema at 93.91%, and a small blister-like lesion near the ear as Contact Dermatitis at 91.13%. On all four uploads, the area of the lesion showed the highest Grad-CAM activation, which is the expected result and an indication of sanity for the vision part of the model. The confidence on the Urticaria test was lower, which is also expected: raised weals are generally less prominent and more transient than plaques or blisters, and their appearance is often more variable, which reflects in the confidence score.

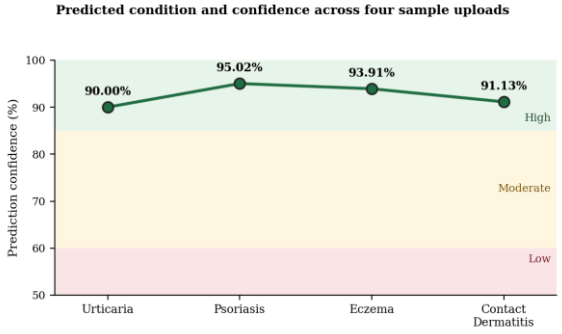


Fig. 2.

Prediction analysis of the diseases

Apart from the test case, it should be noted that the application does not present the user with a disease name and move on. The predicted label always has a visual grounding, a confidence score, an alternative (according to the severity estimation from the symptoms), guidance relevant to the disease and a way to contact a dermatologist. The rule-based elements are kept separate from the model in all cases, making the entire system auditable.

That said, the limitations of the current approach should be acknowledged. Skin appearance is highly variable under different lighting conditions, camera hardware, skin tone, body location and background complexity, and a four-class classifier is not sufficient to cover the spectrum of possible conditions. The recommendations based on the predicted label may be inappropriate for patients with co-occurring conditions, contraindications, allergies or intolerance to medications, and for rare diseases that the model has not been trained to identify. Therefore, the application is to be considered a patient

education and preliminary screening tool, which supplements but does not replaces a medical consultation.

Finally, it is important to note the limitations of the evaluation methodology used. A single run of the model on a test subset that has been prepared by the researchers provides little grounds for statistical inference on the accuracy of the model on photos of varying quality and lighting conditions in the real-world setting. Before a formal release, the same validation procedure must be repeated on a larger test set to verify that the accuracy across classification axes (the four disease labels) is consistent within reasonable margins; preferably, the validation subset must be held out from the training procedure entirely. The Errors should also be assessed with regard to an independent medical A professional's evaluation of the same photos to make sure that the bezel size is appropriate. The errors made by model are not unique or systematic compared to those from human. errors. The informal validation process involved transferring images to the website. The symptoms of skin diseases to application and monitoring the expected The model is trained at the class level for each class image and generates a confidence score and the Grad-CAM visualization for each of them.

A proper validation procedure would involve repeated runs from a fixed test subset, observing the metrics and ensuring consistency from run to run, and reviewing the error cases to assess whether they are reasonable (e.g., two conditions that appear similar on a photo) or not (the model fails to recognize a condition because of an irrelevant detail in the background).

XII. ETHICAL AND PRIVACY CONSIDERATIONS

A medical application that processes patient photos and personally identifiable information must address privacy and ethical concerns at its design stage, rather than retroactively. Images of the skin can be considered sensitive data, and pairing them with a name, age, contact details and location increases the security risks, which is why storing this information on the server is discouraged in favor of retaining it within the session, and generating the PDF report locally. If long-term patient tracking and management is planned, an encrypted database with the patient's consent to store and process their data must be implemented.

It is also worth noting that a medical classifier's predictions can be particularly dangerous if the confidence in them is incorrectly estimated. The interface attempts to alleviate this risk by distinguishing between low, medium and high confidence predictions, using language that discourages the patient from attempting to self-treat ("possible condition") and encourages consultation with a dermatologist. A clinical validation of the same sort will be required before this type of software can be used in a medical institution; ethical review, skin tone bias audit and a regulatory clearance are mandatory for any medical software, AI-assisted or not.

XIII. CONCLUSION AND FUTURE WORK

This paper introduces DermaAI, a deep-learning prototype for screening four common skin conditions, which seeks to go beyond simple classification. EfficientNetV2-B0 is used as a backbone for feature extraction, Grad-CAM is applied to render the classification step interpretable, and rule-based severity engine, personalized recommendations, voice interface and dermatologist resource lookup are added on top of the core vision model. The design choices are organized in a modular fashion around the task of patient triage, resulting in ten distinguishable components summarized in Table II, each of which can be reviewed, tested and modified independently from the others.

The follow-up work will focus on addressing the validation and data collection challenges first, then expanding the system's capabilities in a number of suggested directions. Class-wise statistics on the training and test sets must be computed and published as a part of the evaluation, along with an improved confidence calibration procedure. The model needs to be tested. on a larger sample of data that has better representation across various skin tones and skin devices and dermatologist advice, must be acquired for each validation case to allow for comparison with the The errors made by model compared to human performance. In the longer term, Based on these features, more advanced data augmentation techniques, a mobile-friendly web interface, patient history tracking and a dashboard for visualizing data trends and patterns. recommendation refinement will enable application to be served. To become a more complete dermatology assistant in the future. This is in no way designed to detract from the primary purpose of this project, which was to prove the image classification, Explainable, symptoms reasoning and practical patient Advice may be integrated in one user-friendly application. If future validation efforts confirm the initial assumptions, this approach can be extended to other conditions with visual manifestations beyond the four targeted by this prototype, without significant changes to the severity estimation, recommendation or reporting pipeline.

REFERENCES

[1] Y. B. Nawale and S. A. Gade, "Skin Disease Prediction Using Efficient-Net with Integrated Clinical Decision Support System," International Journal of Engineering Research & Technology (IJERT), vol. 15, no. 1, 2026.

[2] S. S. Han et al., "Augmented Intelligence Dermatology: Deep Neural Networks Empower Medical Professionals in Diagnosing Skin Cancer and Predicting Treatment Options for 134 Skin Disorders," Journal of Investigative Dermatology , vol. 140, no. 9, pp. 1753-1761, 2020.

[3] M. Harangi, "Skin Lesion Classification with Ensembles of Deep Convolutional Neural Networks," Journal of Biomedical Informatics , vol. 86, pp. 25-32, 2018.

[4] A. Esteva et al., "Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks," Nature , vol. 542, no. 7639, pp. 115-118, 2017.

[5] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," Proceedings of the 36th International Conference on Machine Learning (ICML) , 2019.